Explainable Money Laundering Bad Actor Detection

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Team 45

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Canada's Money Laundering Crisis

72.4% of ML cases linked to **drug trafficking** (RCMP)

Cullen Commission: billions laundered through British Columbia alone

Financial institutions must detect bad actors among millions of legitimate customers

Core challenge: finding needles in a haystack with almost no ground truth

$40–100B

laundered annually in Canada

Sources: Canada's 2025 National Risk Assessment, Cullen Commission Final Report, RCMP, CISC 2024

The Data: Scale and Scarcity

61,410
Total Customers

5.9M
Transactions

10
Positive Labels

- 53,000 individuals + 8,000 small businesses
- 7 transaction channels: ABM, Card, Cheque, EFT, EMT, Western Union, Wire
- Only 1,000 customers labeled — of those, just 10 are positive (1%)
- 60,410 customers completely unlabeled — any could be positive
- Card channel alone: 3.55 million rows requiring chunked processing

Solution: 4-Stage Ensemble Architecture

STAGE 1: UNSUPERVISED

Method: Isolation Forest, ECOD, LOF, kNN

20%

Purpose: Statistical anomalies without labels

STAGE 2: DOMAIN RULES

Method: 15 AML rules from Knowledge Library

15%

Purpose: FINTRAC / FATF domain knowledge

STAGE 3: SUPERVISED

Method: XGBoost + LightGBM (averaged)

50%

Purpose: Learn from the 10 labeled positives

STAGE 4: PU LEARNING

Method: Retrained XGBoost + LightGBM

15%

Purpose: Expand training with reliable negatives

$$\text{risk_score} = 0.20 \times \text{unsupervised} + 0.15 \times \text{rules} + 0.50 \times \text{supervised} + 0.15 \times \text{PU}$$

AML Knowledge Library

30
Red Flags · 8 Typologies

Money Mule: students/young people receiving frequent transfers

Structuring: splitting transactions to avoid $10K CAD threshold

Shell Company: holding companies with zero sales/employees

Layering: rapid fund movements across multiple channels

Income Mismatch: activity grossly exceeding declared income

Cash-Intensive: frequent large ABM cash transactions

High-Risk Industry: holding, real estate, legal, financial, MSB

Organized Crime: combined indicators from CISC/RCMP typologies

Sources: FINTRAC operational alerts, FATF typology reports, Cullen Commission, CISC 2024 organized crime assessment, Canada's 2025 NRA, RCMP case analysis

Feature Engineering: 229 Domain-Informed Features

KYC Static (~40): age, tenure, occupation flags, income, industry flags, employee count

Per-Channel Transaction (175): volume, amounts, AML signals, temporal

Cross-Channel Behavioral (14): active channels, HHI, income ratios, structuring

Feature Spotlight & AML Indicator

transaction_to_income_ratio	Activity inconsistent with financial profile	FINTRAC Cat. 3
total_near_10k	Structuring below $10K reporting threshold	PCMLTFA
industry_is_holding + zero_sales	Shell company with no economic purpose	FATF 2018
channel_hhi (low)	Layering across channels to obscure trail	FINTRAC Cat. 6
abm_cash_count	Cash-intensive activity	FINTRAC 2018

Domain Rule Engine: 15 AML Rules

- Each rule encodes a specific FINTRAC/FATF money laundering typology
- Weighted by severity: HIGH (1.0), MEDIUM (0.5), LOW (0.25)
- Produces human-readable descriptions for every triggered rule

Money Mule

Student/unemployed + >20 incoming EMTs

SEVERITY: HIGH (1.0)

Structuring

>3 transactions in $9K–$10K range

SEVERITY: HIGH (1.0)

Shell Company

Holding company + zero sales or employees

SEVERITY: HIGH (1.0)

Income Mismatch

Transaction volume exceeds incorne by 50x+

SEVERITY: HIGH (1.0)

Layering

5+ channels, low concentration, high volume

SEVERITY: HIGH (1.0)

High-Risk Industry

Holding, real estate, legal, financial, MSB

SEVERITY: MEDIUM (0.5)

* Coverage on known positives: 7/10 income mismatch • 4/10 high-risk industry • 4/10 layering

Supervised + PU Learning

Stage 3: Supervised (50%)

- XGBoost + LightGBM (averaged)
- Shallow trees: max_depth = 4
- scale_pos_weight = 99 (class imbalance)
- Anomaly + rule scores as features (stacking)
- LOPO validation: hold out each positive, retrain, confirm detected

Stage 4: PU Learning (15%)

- Use initial classifiers to score unlabeled customers
- Identify reliable negatives (predicted prob < 0.1)
- Retrain with expanded labeled set
- 2 iterations of refinement
- Dynamically adjusts class weights

Key Insight

With only **10 positives**, supervised learning alone would overfit.
The ensemble lets each stage compensate for the others' weaknesses.

Explainability: Every Score Tells a Story

Three-Layer Explanation Approach

- **Triggered AML Rules:** human-readable descriptions citing FINTRAC/FATF
- **Top 5 SHAP Features:** XGBoost feature importances translated to plain English
- **Risk-Tier Templates:** depth of explanation scales with risk level

HIGH RISK (score: 0.87)

This customer has an elevated risk score.
Indicators identified:

- Transaction volume exceeds income by 312x
- High incoming e-transfers (student)
- Layering across 5+ channels

LOW RISK (score: 0.02)

This customer has been assessed as low risk.
Transaction activity is consistent with the customer's profile.

No significant AML red flags were identified.

Results

10/10
Positives Detected

Top 40
out of 61,410

3.0%
Flagging Rate

10/10
LOPO Valid.

Metric	Value
Known positives flagged (predicted_label = 1)	10 / 10 (100%)
Worst positive rank by risk_score	40th out of 61,410
Flagging rate	3.0% (~1,843 customers)
Leave-One-Positive-Out validation	10 / 10 pass
Threshold	**97th percentile of risk_score**

LOPO: hold out each positive one at a time, retrain the full pipeline, confirm it is still detected

Bad Actor Archetypes

Patterns discovered among the 10 known positives:

Money Mules

Students with $8 declared income receiving thousands in wire transfers and incoming e-transfers

Shell Companies

Brand-new holding / law firm with $0 sales, minimal employees, yet high transaction volume

Layering Operators

Real estate businesses using 5+ channels with low concentration, rapid fund movements

Income Mismatch

Transaction volume exceeding declared income by 50–300x, inconsistent with customer profile

Each archetype is caught by different ensemble stages — no single method detects all patterns

Conclusion & Future Work

Three Key Differentiators

Domain-Grounded: 30 red flags sourced from FINTRAC, FATF, Cullen Commission, CISC, RCMP

Hybrid Architecture: unsupervised + rules + supervised + PU learning handles extreme scarcity

Full Explainability: SHAP + rules + tiered templates produce actionable explanations for all 61,410 customers

Future Directions

Temporal modeling: transaction sequences and time-series anomaly detection

Graph-based features: network analysis of fund flows between customers

Active learning: use model uncertainty to prioritize investigation of unlabeled customers

Production deployment: real-time scoring with streaming transaction data

From 10 labels to 61,410 actionable risk assessments